Hossein Shahabadi Farahani

AI Systems Engineer · Solutions Architect

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Iran · Open to remote roles and freelance projects · English: professional working proficiency

PROFILE

AI systems engineer bridging research and working software: RAG pipelines, cloud inference and edge applications. Experience spans industrial fault detection, Persian speech systems and evaluation under hardware constraints. My manifesto describes how I work: systems thinking, intellectual honesty and clear reasoning.

TECHNICAL KNOWLEDGE & SKILLS

Theoretical foundations: Transfer learning & domain adaptation · adversarial training · probabilistic graphical models · MCMC / Gibbs sampling · Boltzmann machines · variational autoencoders · Transformers · U-Net · image segmentation · Grad-CAM · embeddings · hybrid retrieval · RAG

AI tools & frameworks: PyTorch · TensorFlow · Hugging Face · Ollama · LangChain · vector databases (Chroma) · model serving

Backend, web & software development: Python · FastAPI · REST APIs · Docker · Linux · Git · pytest · Poetry · GitHub Actions · D-Bus · Raspberry Pi

EXPERIENCE

Machine Learning / MLOps Engineer

MAPNA Digital / MECO, MAPNA Group 2022 - Present

- **Persian voice assistant:** Designed and implemented an end-to-end smart-home prototype: a 1 GB Raspberry Pi client, four containerized cloud services and D-Bus device integration. Case study
- Built per-user intent classifiers with a frozen ParsBERT encoder and **<40 KB trained heads**; used LLM-generated training examples at setup, keeping LLM calls out of inference.
- Observed **96% command recognition** and **2.85 s mean request-to-action-ID latency** in small prototype trials (3-4 speakers, roughly 10 utterances each; low network latency).
- Deployed speech-to-text and RAG services with **FastAPI and Docker**; integrated classification, retrieval and model storage. Built a chest X-ray classification proof of concept with Grad-CAM explanations.

AI Strategy & Technical Pre-Sales (concurrent responsibilities)

MAPNA Digital / MECO, MAPNA Group 2021 - Present

- Assessed AI feasibility, defined validation criteria and presented technical solutions to international visitors in English. Designed a PRD and Confluence workflow for a roughly 100-person R&D division; implementation was partial.

Data Scientist, Condition Monitoring

MECO, MAPNA Group · APAC Research Group, KNTU 2018 - 2020

- Built anomaly detection and root-cause analysis across hundreds of power-plant sensor streams; tools supported diagnosis of failures in operating plants.
- Developed domain-adaptive soft sensors for process-variable estimation across units; research published in *Computers in Industry*.

SELECTED OPEN-SOURCE PROJECT

Offline RAG & Evaluation

Independent project · Code · Case study 2026

- Adapted five DeepLearning.AI RAG assignments to **Ollama and Chroma**, running locally on a CPU-only laptop; added pytest regression coverage and GitHub Actions CI.
- Replaced LLM routing with embedding similarity: **161 s to 36 s median latency**, with correct answers rising from **4/12 to 8/12** on a fixed, 12-question benchmark using a 1.5B model.

EDUCATION

MSc, Mechatronics Engineering · K. N. Toosi University of Technology

BSc, Mechanical Engineering · University of Arak

PUBLICATIONS

- **Domain Adversarial Neural Network Regression to Design Transferable Soft Sensor in a Power Plant.** *Computers in Industry* (Q1, IF: 10), 2021.
- **On the Application of Domain Adversarial Neural Network to Fault Detection and Isolation in Power Plants.** *19th IEEE ICMLA*, 2020. IEEE Xplore
- **Transfer Learning in Condition Monitoring of Process Plants.** *2nd BRIC Symposium*, McMaster University, 2022.
- **Between-Domain Instance Transition via Gibbs Sampling.** *arXiv*, 2020.